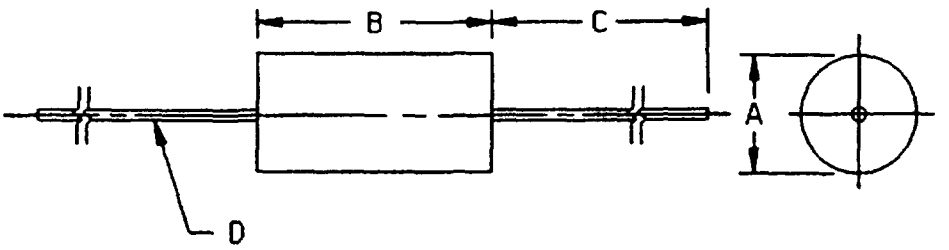
INCH-POUND
MIL-C-39010/1D
4 January 1994
SUPERSEDING
MIL-C-39010/1C
1 March 1985

MILITARY SPECIFICATION SHEET

COILS, FIXED, RADIOFREQUENCY, MOLDED, MICROMINIATURE, MAGNETICALLY SHIELDED (PHENOLIC CORE-IRON SLEEVE), ESTABLISHED RELIABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-39010



Ltr	Dimensions are in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	152 (3.86)	172 (4.37)
B	.390 (9.91)	.430 (10.92)
C	1.250 (31.75)	1 626 (41 30)
D	023 (0.58)	.027 (0.69)

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for information only

FIGURE 1 Dimensions and configuration

(D) denotes changes

AMSC N/A FSC 5950

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REQUIREMENTS:

Design, construction, and physical dimensions. See figure 1

Material Phenolic core with iron sleeve

(D) Weight 0.03 ounce maximum

Operating temperature range -55°C to +105°C.

Dielectric withstanding voltage Method 301 of MIL-STD-202; test voltage of 1,000 V rms.

Barometric pressure Method 105, test condition C, of MIL-STD-202, 70,000 feet with a test voltage of 200 V rms.

Electrical characteristics See tables I and II.

Inductance See table I

Inductance tolerance See table I

Q values See table I

Self-resonant frequency See table I.

Percent coupling 3 percent maximum

DC resistance See table I.

Temperature rise 15°C

Terminal pull 5 pounds

(D) Part or Identifying Number (PIN) M39010/01 (dash number from table I)

(D) Supersession data This specification supersedes MS75087 dated 4 September 1985

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TABLE I. Electrical characteristics (initial) and dash numbers

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q minimum	Test frequency (MHz)	Self-resonant frequency minimum (MHz)	DC resistance (25°C) maximum (ohms)	Rated dc current (mA) 2/
AR10**	.10	5, 10	50	25	250	.025	1,790
AR11**	11	5	50	25	250	.034	1,530
AR12**	.12	5, 10	50	25	250	.034	1,530
AR13**	13	5	50	25	250	.037	1,470
AR15**	.15	5, 10	50	25	250	.037	1,470
AR16**	.16	5	50	25	250	047	1,300
AR18**	18	5, 10	50	25	250	047	1,300
AR20**	.20	5	49	25	250	.067	1,100
AR22**	.22	5, 10	49	25	250	067	1,100
AR24**	24	5	47	25	250	11	853
AR27**	.27	5, 10	47	25	250	11	853
AR30**	.30	5	46	25	250	13	780
AR33**	.33	5, 10	46	25	250	13	780
AR36**	.36	5	44	25	250	18	670
AR39**	.39	5, 10	44	25	250	18	670
AR43**	.43	5	44	25	235	25	565
AR47**	47	5, 10	44	25	235	.25	565
AR51**	.51	5	43	25	210	33	490
AR56**	.56	5, 10	43	25	210	33	490
AR62**	62	5	42	25	190	.45	420
AR68**	68	5, 10	42	25	190	45	420
AR75**	.75	5	40	25	180	.59	370
AR82**	82	5, 10	40	25	180	59	370

1/ The complete dash number will include two additional letters (indicated by **). The first additional letter will indicate the inductance tolerance and the second additional letter will indicate the failure rate symbol (e.g , M, P, R, S) and will be added to the end of the dash number

2/ The rated dc current is based on 90°C ambient temperature with a 15°C rise

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(D) TABLE II Electrical characteristics (final) 1/

Inspection group	Allowable variation from initial measurement		Allowable percent from specified minimum value in electrical characteristics (initial) table	
	Inductance (percent)	DC resistance	Self-resonant frequency	Q
Qualification inspection				
Group II	±5	---	---	-10
Group IV	±5	±(3% + 001 ohm)	2/	-10
Group V	±5	±(2% +.001 ohm)	2/	-10
Quality conformance inspection Group B				
Subgroup 1	±5	±(2% + 001 ohm)	2/	-10
Subgroup 3	±5	---	---	-10
Subgroup 4	±5	±(3% + 001 ohm)	2/	-10

1/ Test fixture allowance of + 01 µH shall be added to all change in inductance limits ±(_ percent + 01 µH).

2/ The self-resonant frequency shall be not less than the value specified in table I

CONCLUDING MATERIAL

Custodians.
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AR, ME, MI
Navy - AS, CG, MC, OS, SH
Air Force - 17, 19, 99
DLA - ES

Preparing activity
Army - ER

Agent
DLA - ES

(Project 5950-0831-01)

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